



Curiosity update

NASA's Curiosity rover on Mars has obtained samples containing nitrates that are necessary for life to exist <http://dgit.in/CurNit>



Acer Revo One RL85

Seen first at CES this year, the Revo One RL85 from Acer is a tiny PC primed for home entertainment <http://dgit.in/RevoOne>

VR Headsets: Blurring the line between fantasy and reality

Virtual Reality shows real promise with the devices that are currently under development. Let's take a look at the ones that are betting big.

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Man is forever searching for ways to enhance his perception of reality and escape inherent physical, economical, and social limitations. Who among us would not want to experience the rush of adrenaline when doing a barrel roll in a fighter jet, the sense of trepidation when spelunking about uncharted caves in search of ancient relics, or the heightened feeling of unparalleled awe when floating in outer space? With the magic of virtual reality, such scenarios are slowly becoming a possibility in the near future.

Much of the credit must be given to pioneering devices such as the Oculus Rift, which with its complete visual immersion, 360-degree head-tracking, and comprehensive open-source support, brilliantly demonstrated the enormous potential of VR headsets in gaming and multimedia. Although the Oculus Rift did enjoy most of the acclaim for being the first device to bring VR to mainstream technology, bigger players such as Sony, Samsung, HTC, and Microsoft soon realized the potential of this nascent industry and have now introduced similar devices to compete for sales in the VR device market. Furthermore, with Facebook recently acquiring Oculus for



The future of gaming

a whopping \$2 billion, other companies will be forced to increase funding for their own VR devices, a win-win situation for the consumer.

Virtual Reality Headsets 101

Mainstream VR headsets follow a basic design standpoint: a hard plastic body, which is used to house the motherboard and processor, display, viewing lenses, and in some cases, an audio module and a camera. All these components are interfaced with a secondary device such as a PC or smartphone, which is used to process the game or virtual environment. Coupled with advanced features such as 360-degree head tracking, sensors such as gyroscopes and accelerometers, and high-definition 3D-audio, the final product is a device capable of generating fully immersive 3D environments.

While each VR headset does have its pros and cons, there are some common attributes that determine the overall VR experience. Parameters such as display type, screen resolution, refresh rate (or maximum framerate), field of view (FOV), and input latency can vary immersiveness and responsiveness, while design, build quality and ergonomics can influence comfort and usability. Of course, there's only so much that can be summarized in such a short explanation. Let's take a more detailed look at the devices that might soon become electronic mainstays in our households.

The Oculus Rift

Ever since its humble beginnings as a Kickstarter campaign, the Oculus Rift has been synonymous with VR headsets. The device has come a long way since